WasteSort Tuesday Speaking Guide

Goal Diggers | Day 2: dataset inventory and preprocessing | About 4-5 minutes

1. Preparing the WasteSort dataset

Good morning. We are Goal Diggers. For Tuesday, our task is to prepare the dataset before model training. We load image paths and labels in Python and Pandas, inspect image formats and duplicates, and define resizing and label encoding. The outputs are a dataset inventory and a preprocessing notebook that the group can run and explain.

2. What the inventory records

Each row in the inventory represents one source image. We record its file path, material label, numerical label code, format, dimensions and whether it is included. Pandas lets us count and inspect these records without confusing the table with the image itself. The folder supplies the original label. Excluded files stay in the inventory with a reason, so the cleaning decisions remain traceable.

3. Class counts before and after cleaning

The source contains 2,527 images in six categories. Our automated checks exclude six files, leaving 2,521 usable images. Paper is the largest class and trash is the smallest. This imbalance matters because a model can achieve a reasonable overall accuracy while missing a small category. These counts describe our data preparation, not model performance.

4. Duplicate checks prevent misleading evaluation

We found three pairs with identical decoded pixels but conflicting material labels. Here, the same pixels appear in both glass and metal folders. We excluded both copies instead of guessing which label was correct. We also found 45 possible near-duplicate pairs using perceptual hashes. These are candidates for human review. Connected candidates stay together in one partition to reduce detected leakage.

5. Separate training, validation and test images

The usable images are divided into 1,774 training, 371 validation and 376 test images. Training fits the model. Validation helps choose settings and the model. Test images support final evaluation after that choice is frozen. We use seed 42 and preserve candidate groups. The proportions are approximately 70, 15 and 15 percent; the actual counts are saved so the split can be reproduced.

6. Labels become category identifiers

We encode the labels consistently: cardboard is zero, glass one, metal two, paper three, plastic four and trash five. These numbers are identifiers only. A label of five does not mean five units or a material ranked above four. This remains a classification problem. The same mapping must be used during training, evaluation and prediction so the app displays the correct class name.

7. Preprocessing depends on the model

For the classical models, we resize to 64 by 64 grayscale pixels and calculate 1,764 HOG features. For neural models, we resize to 160 by 160 RGB pixels. The small CNN uses values between zero and one; MobileNetV2 also uses its pretrained normalization. Square resizing changes the original aspect ratio, which is a baseline tradeoff. Random augmentation is applied only to training images.

8. Tuesday handoff

The Tuesday handoff includes the complete inventory, usable-image manifest, exclusions, encoding, preprocessing plan and notebook. The notebook checks the saved split and demonstrates input batches without training a classifier. A review worksheet is included for the group to inspect labels and possible duplicates. Automated preparation is complete, but human review remains a separate responsibility. These prepared inputs support the next model-training activity.

Notebook Demo and Practice Questions

A short demonstration

1. Open WasteSort_Day2_Tuesday.ipynb in Google Colab. A CPU runtime is enough. Select Runtime > Run all and wait for the cells to finish.
2. Show the inventory table: explain path, label, label_id, format, dimensions and inclusion status.
3. Show class counts and exclusions: 2,527 original images, six excluded files and 2,521 usable images.
4. Show the saved split and successful checksum check. Explain training versus validation versus test.
5. Show the three input shapes and the resize comparison. Download the Day 2 export ZIP.

If internet access is unreliable

Use the executed notebook, the presentation PDF and the included CSV files. Say that these are saved outputs from the documented local execution; do not describe them as a live Colab run.

What is the inventory?

A table with one row per source image, including its path, label, properties, inclusion status and partition.

Why keep excluded files in the inventory?

To document what was removed and why. The usable-image manifest alone would hide those decisions.

Why exclude both conflicting copies?

Identical pixels have conflicting labels. Without verified ground truth, choosing one label would be an unsupported correction.

Are all 45 candidate pairs true duplicates?

No. Perceptual hashes screen for visual similarity. A person still needs to inspect them.

Why keep candidate groups together?

Related images in different partitions could make evaluation look better than generalization to genuinely new objects.

Does encoding make classification into regression?

No. The integers are class identifiers. Their numerical differences have no material meaning.

Why use different image sizes?

Different model families use different input representations. Each model must use the same preprocessing in training and prediction.

What still needs review?

Human label and candidate-pair inspection, independent reproduction and explanation by the group, and instructor review of the revised protocol.

Evidence and sources

The supplied Tuesday task; dataset_inventory.csv, data_audit.json, excluded.csv, split_counts.csv, label_encoding.json, preprocessing.json and protocol.json in the project repository. TrashNet by Gary Thung and Mindy Yang.

AI assisted preparation and execution. Group review and actual member contributions must be recorded separately.